

The same two curves, four times

One picture, four wirings, and **less of my working shown each time**. We do items 1 and 2 in the room — nine minutes, and item 1 is already done. Items 3 and 4 are on **PS3**.

The procedure, every time:

1	Draw the production rate against p .	3	Mark every crossing.
2	Draw the removal rate against p on the same axes.	4	Classify each one by the <i>sign of the gap</i> either side.

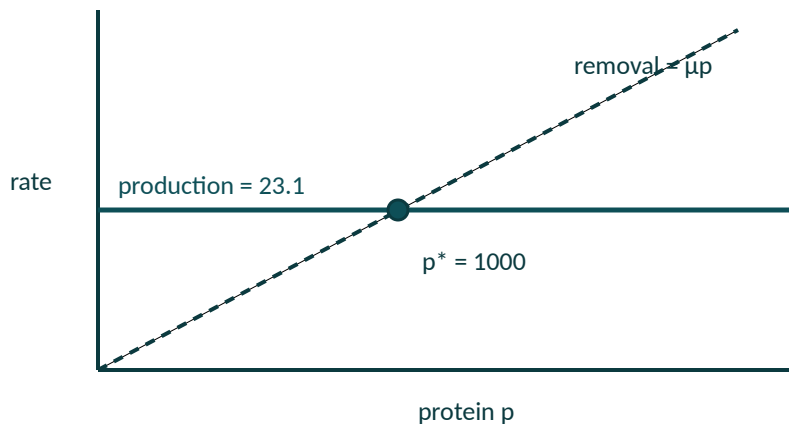
Throughout, one host and one target: a protein that must sit at $p^* = 1000$ molecules in *E. coli* dividing every 30 minutes, so $\mu = \ln 2/30 = 0.0231 \text{ min}^{-1}$. Unless a tag is added, $\gamma = 0$ and removal is dilution alone. The unregulated promoter that hits the target is therefore $\alpha_0 = \mu p^* = 23.1$ proteins per minute.

1 • Fully worked — the constitutive unit

No feedback, so production does not depend on p :

$$\frac{dp}{dt} = \underbrace{\alpha_0}_{\text{production}} - \underbrace{\mu p}_{\text{removal}}$$

Draw them. Production is a horizontal line at **23.1**. Removal is a straight line through the origin of slope μ — a *fraction* of the pool leaves per minute, so the amount leaving grows with the pool.



Find the crossing. $\alpha_0 = \mu p^*$, so $p^* = \alpha_0 / \mu = 1000$. That is the only thing the crossing tells you, and it is the *level*.

Show it is stable, without solving anything. To the right of the crossing removal is above production, so $dp/dt < 0$ and p falls back. To the left, production is above removal, so p rises. The gap **changes sign** through the crossing, from positive to negative — that is what stable means, and notice the argument never used the functional form of either curve.

► **Why does that matter?** Name one thing you could change about the production curve that would leave this stability argument untouched.

2 · The last step is yours — negative autoregulation

Now the protein represses its own promoter, so production falls as p rises:

$$\frac{dp}{dt} = \frac{\alpha}{1 + p/K} - \mu p$$

Take $K = p^* = 1000$, which is a **repression ratio of 2** — a modest circuit.

Draw both curves on these axes. Removal first, then production.



► **Fix the level.** Repression pulls the production curve down, so the crossing moves left and the cell misses its target. Choose α so the crossing sits back at $p^* = 1000$. One line:

$$\alpha =$$

► **Which curve moved, and which did not?** Answer for the production curve and for the removal curve separately, and say what that means for the rate of synthesis **at the crossing**.

► **Why is it faster?** Compare the vertical gap between the two curves at $p = p^*/2$ in item 1 and in item 2. Which is larger, and what does the gap *mean*?

2 continued · two checks, and do both — they are worth more than the drawing

► Let $K \rightarrow \infty$. You should recover item 1 exactly. Say why in words, not algebra.

► Now make the repression aggressive: $K = p^*/10$. What happens to α , and what happens to the synthesis rate at the crossing? One of those two changes and one does not, and that is the entire cost result.

3 · On PS3 — positive autoregulation

The protein now **activates** its own promoter, with n sites bound cooperatively and a leaky basal rate:

$$\frac{dp}{dt} = \alpha_{\text{basal}} + \alpha_{\text{max}} \frac{p^n}{K^n + p^n} - \mu p$$

Find every crossing and classify each one. Then find the value of n at which the middle pair appears, and say what happens to a cell sitting exactly at the unstable crossing.

And the question that matters: μ is the *slope of the removal line*. What happens to your answer when the same construct is grown in minimal medium?

4 · On PS3 — the design item

You are handed a protein that must reach p^* in **under 15 minutes** and then hold that level for **a day**. You may add a degradation tag, wire negative autoregulation, or both.

Pick a route, **price it** — proteins per minute, integrated over the day — and defend the choice. Then say what you would measure to know whether the circuit you built is the circuit you designed.